Predicting Earthquake Damage Grade from Building Attributes: Machine Learning on 762,094 Buildings from the 2015 Gorkha (Nepal) Earthquake

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Objective

Test how well cheap, pre-earthquake building attributes (structural system, wall and foundation materials, storeys, age, plan) predict the observed damage grade (1 = slight cracking ... 5 = collapse), and check whether the learned patterns agree with structural-engineering knowledge, as a step towards data-driven rapid seismic vulnerability screening.

Data

Building-structure table of the Government of Nepal *Household Registration for Housing Reconstruction* survey (National Planning Commission / Central Bureau of Statistics; 2015 Nepal Earthquake Open Data Portal, eq2015.npc.gov.np, built by Kathmandu Living Labs). 762,106 buildings in 11 districts; the identical CSV was obtained from a public GitHub mirror because the official portal was unreachable, and verified by row count and SHA-256. Four post-earthquake columns were removed to prevent target leakage; 12 rows without a label were dropped (n = 762,094); age = 999 was treated as unknown. 22 building features were used. No explicit data license is published; used for non-commercial research with attribution.

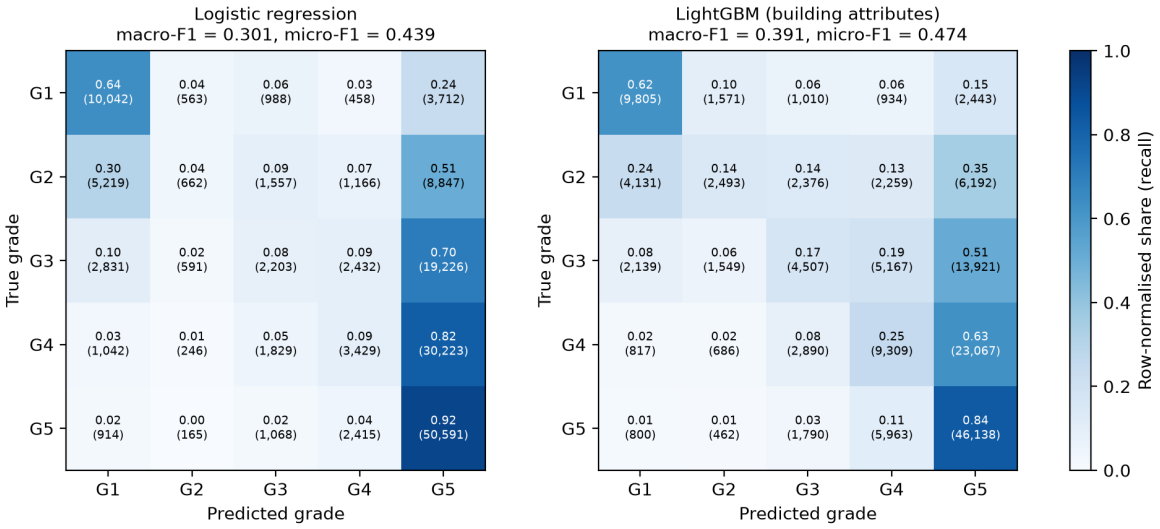
Method

Stratified 80/20 split (609,675 train / 152,419 test). Baseline: multinomial logistic regression (stratified 200k training subsample). Main model: LightGBM with native categorical handling and early stopping (742 trees). Ablation: LightGBM + district (location proxy for shaking). Stratified 5-fold CV on a 100,000-row training subsample. Explanations: SHAP (TreeExplainer) and permutation importance. Total CPU runtime approx. 4 min.

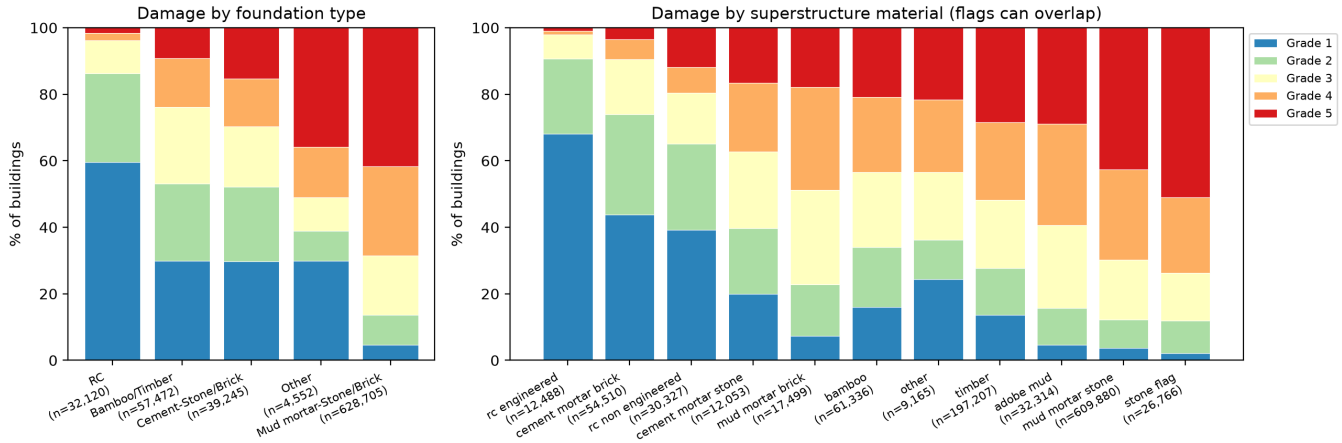
Results (held-out test set)

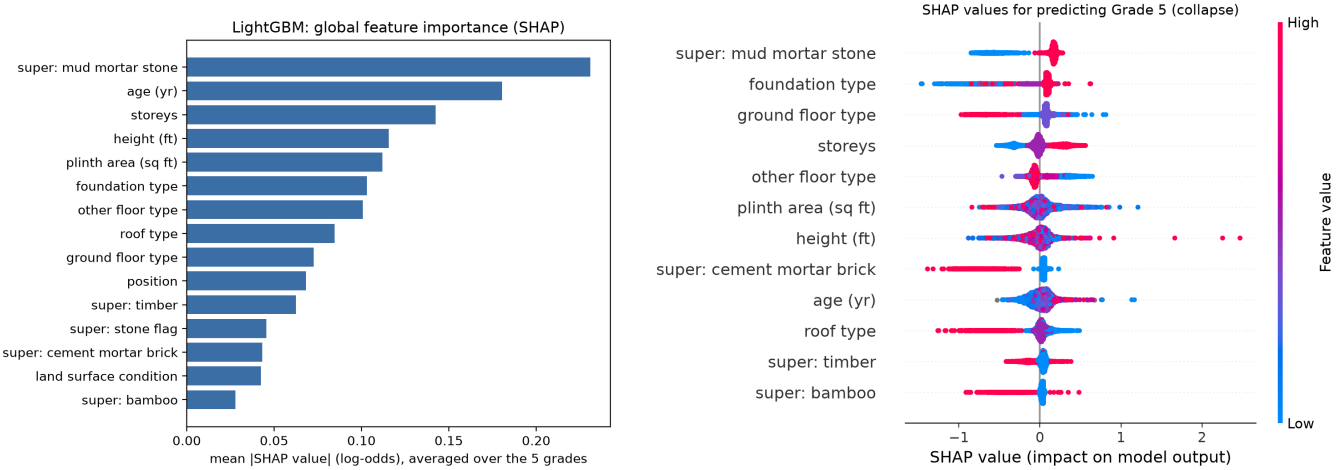
Model	Macro-F1	Micro-F1 (acc.)	Within +/-1 grade
Majority class (always Grade 5)	0.106	0.362	0.603
Logistic regression	0.301	0.439	0.733
LightGBM, building attributes only	0.391	0.474	0.781
LightGBM + district (ablation)	0.472	0.529	0.844

5-fold CV macro-F1: logistic regression 0.299 ± 0.002, LightGBM 0.371 ± 0.002. LightGBM recognises collapse (Grade 5 recall 0.84) and slight damage (Grade 1 recall 0.62) well; intermediate grades 2-4 are often confused with neighbours. Building attributes alone cannot see ground-shaking intensity; adding only the district raises macro-F1 substantially, showing that hazard information is the main missing ingredient.



What the model learned: engineering interpretation





Mud-mortar stone masonry is the strongest predictor in both SHAP and permutation importance (mean Grade-5 SHAP +0.17 log-odds when present). Rubble stone in mud mortar has negligible tensile/shear strength and heavy walls without bands or through-stones, so walls delaminate and collapse out-of-plane; 42.7 % of such buildings collapsed. **Engineered RC** (-0.97), **cement-mortar brick** (-0.76) and **RC roof slabs** (-0.64) push predictions towards low damage, consistent with rigid diaphragms, mortar bond and a defined lateral-load path; **non-engineered RC** helps far less (-0.25). Age, storeys and height follow next (deterioration, pre-code construction, larger overturning demand on weak masonry). These are model explanations, not causal proof: materials are also correlated with location and shaking intensity.

Relevance to India

Unreinforced stone/brick masonry, timber-bamboo and non-engineered RC houses, the typologies driving damage here, form a large share of Indian housing, especially in the Himalayan states and Indo-Gangetic plain. Under IS 1893 (Part 1):2016 Lucknow is in seismic Zone III, and the revised IS 1893 (Part 1):2025 (Annex D, Table 18) also lists Lucknow in Zone III (Z = 0.25, 2,475-year return period). A model of this type, retrained or recalibrated on Indian damage surveys (e.g. Bhuj 2001, Sikkim 2011, Imphal 2016) and combined with the zone factor or site-specific ground motion, could rank buildings from census / property-tax attributes for detailed evaluation and retrofit (IS 13935), complementing rapid visual screening.

Limitations and next steps

Single earthquake, region and survey; no per-building ground-motion input; intermediate grades hard to separate (macro-F1 0.39); subsampling used for the logistic baseline and CV; light hyper-parameter tuning; possible label bias from a grant-eligibility survey; dataset license not explicitly stated. Not for decisions on individual buildings. Next: add USGS ShakeMap intensity per ward, ordinal/cost-sensitive models, spatial cross-validation by district, and transfer tests on Indian data. Code, executed notebook and all metrics are in the accompanying project files (see README).